The P2P network simulation

`network_sim.py` puts the coding schemes into a simulated peer-to-peer network and measures what happens to them there. It is the Phase 3 half of the project: where `benchmark_1D_2D_BC.py` asks *what can this code survive*, this asks *can a node actually get the data and rebuild the blob before the slot ends*.

Those are genuinely different questions, and the second one cannot be answered by the first. A decode time of 40 seconds is a fact about arithmetic. Whether 40 seconds is acceptable is a fact about a system with a 12-second slot, a deadline, a finite link and peers that may not answer.

What the project brief asks for, and where it now stands

Phase 3 of the proposal specifies prototypes "deployed within a controlled simulated peer-to-peer (P2P) network environment", evaluated "across standard network architectures", capturing "bandwidth saturation, packet propagation latency, and block processing throughput" under "varying adversarial blockchain network conditions".

Requirement	Status
Simulated P2P environment	done — SimPy discrete-event, 64 peers, custody assignment
Packet propagation latency	done — per-peer lognormal latency, reported as gather time
Bandwidth saturation	done — one contended downlink; saturates under multi-blob load
Block processing throughput	done — blobs reconstructed per slot
Varying adversarial conditions	done — swept 0–80% withholding peers
Randomly generated blob data	done — real EIP-4844 packing, shared blob
Data reconstruction success rate	done — per-trial, byte-verified
Scalability in node count	done — peer count is a swept dimension
Standard network architectures (plural)	**not done — one topology only**

The remaining gap is listed again at the end rather than buried. Everything above it is measured.

How SimPy works, in the three concepts this uses

SimPy is a discrete-event simulator. Nothing in it runs in real time.

The environment is a clock and an event queue. `env.now` is the current *simulated* time in seconds. `env.run(until=4.0)` processes every scheduled event up to simulated t=4s and stops.

A process is an ordinary Python generator. `yield env.timeout(0.08)` means "resume me 80 simulated milliseconds from now". The scheduler jumps straight to the next event rather than waiting, so simulating a 12-second slot across 64 peers costs microseconds of real CPU. This is what makes sweeping hundreds of network configurations cheap.

A resource models contention. The node's downlink is a `simpy.Resource` of capacity 1: a transfer acquires it, holds it for `bytes / bandwidth` seconds, releases it. Because every transfer queues for the same object, bandwidth saturation *emerges* from demand rather than being asserted — ask for more than the link can carry before the deadline and the queue backs up until the stragglers miss it.

The one idea the design rests on

In the coding benchmark, the erasure mask is an **input**: `random_mask(rate, rng)` invents one at a nominal rate. Here it is an **output** — whatever the node failed to collect in time.

```
benchmark:    mask = adapter.random_mask(rate, rng)     # invented
network sim:  mask = <what did not arrive before the deadline>
```

Everything downstream of the mask — decode, extraction, byte-level verification — is the *same adapter code* the benchmark already uses, untouched. That single substitution is the entire integration surface, which is why this file contains no coding logic of its own. A fix to `2D_RS_Galois.py` changes these results automatically.

This buys something beyond convenience. The benchmark drops symbols independently; real failures are not independent, because when a peer goes offline *everything it custodied* vanishes at once. A peer-failure model produces those **clustered** erasures for free, from the topology, rather than having to inject correlation by hand — and the benchmark report explicitly flags i.i.d.-only testing as a limitation.

What is modelled

Peers and custody. 64 peers. Each cell is held by `custody_replication` of them (default 4), so a cell is lost only if every custodian fails. This is the *network's* redundancy, and it is separate from the code's — an important distinction, because it turns out to absorb most peer failure before the erasure code is asked to do anything.

Cells. The unit a node requests is a PeerDAS-shaped cell: 64 field elements × 32 bytes, plus a 48-byte KZG proof, so 2096 bytes.

Adversaries. A fraction of peers never answer. The client burns its request timeout before falling back to another custodian, so withholding costs the victim *time* as well as data — which is the point of modelling it rather than just deleting symbols.

The deadline. A 4-second sampling window inside a 12-second slot. Anything not received by then is an erasure, full stop.

How much a node actually has to download

This is the part worth dwelling on, because getting it wrong produced nonsense on the first attempt.

The obvious answer is k symbols — the information-theoretic floor. That is right only for an MDS code decoded optimally. In general a node must fetch enough that what remains *missing* falls inside its own decoder's tolerance:

> symbols needed $= n \times (1 - \text{tolerance})$

	tolerance used	must fetch	of
1D-RS	0.49	4178	8192
2D-RS	0.38	5238	8448
BC ω1024 λ2	0.33	5489	8192

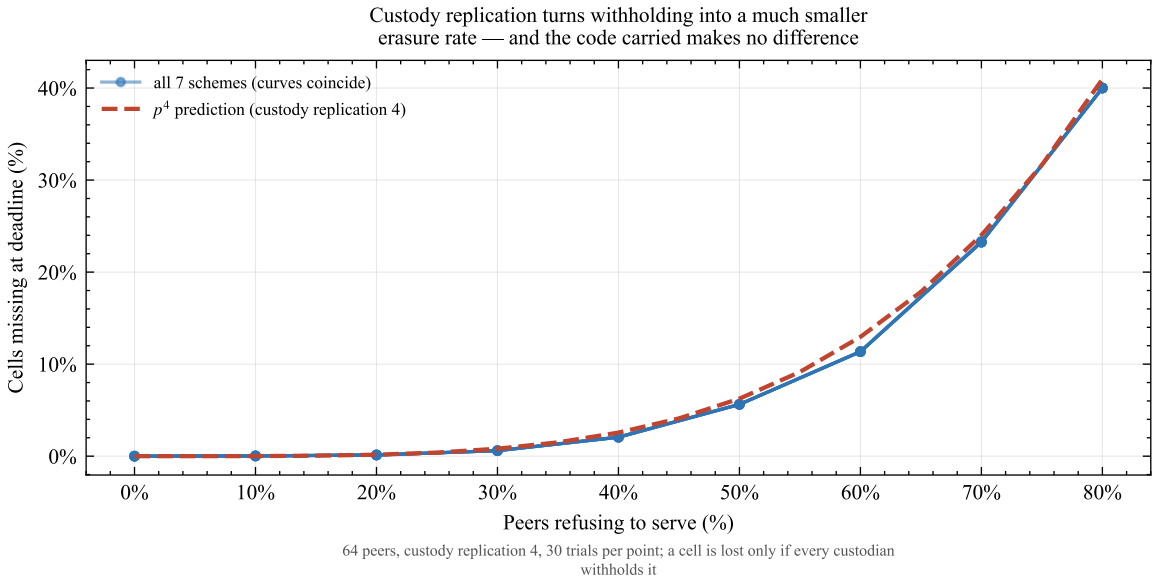
The tolerance here is deliberately **conservative**: the highest swept rate at which *every* trial recovered, not the 50% threshold the benchmark report quotes. A node sizing its download wants to recover reliably, not half the time, so it plans against the point where recovery is certain. That is why these sit below the thresholds of 0.499, 0.400 and 0.342 — the two are different quantities answering different questions, and both are correct. For 1D-RS the rule asks for 2% more than the 4096 symbols an MDS code strictly needs, because it is applied the same way to every scheme, from measured behaviour rather than from each code's structure.

So a weaker decoder forces a **larger download at identical storage cost**. This is the benchmark report's "surviving data needed" figure, no longer sitting in a table but actually driving a network transfer. Setting the fetch target from k alone instead made every scheme sit at 42% erasure even with a perfect network, and BC then "failed" at zero adversaries — an artifact of the fetch policy, not a property of the code.

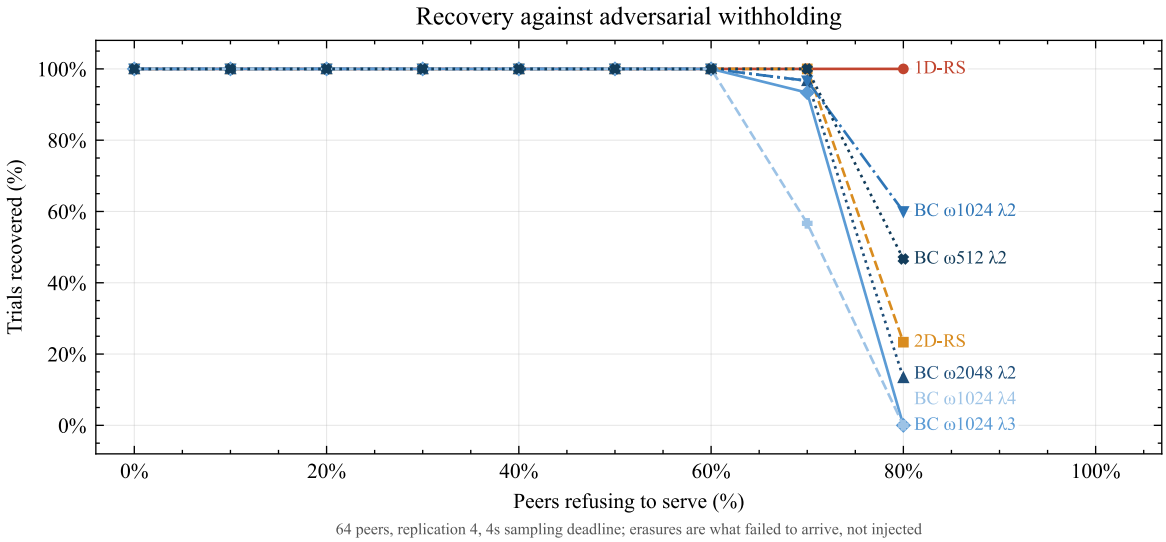
Results

The figures below come from three sweeps, all with 30 trials per point: **A** — one blob at 50 Mbps, all seven schemes, withholding swept 0–80%; **B** — six blobs at 25 Mbps, the same schemes, 0–60%; **C** — peer count 8→256 at 30% and 60% withholding. Every trial is decoded for real (`--real-decode`): the actual decoder runs on the erasure pattern the network produced, and a trial counts only if the bytes match. Raw rows are in the accompanying JSON.

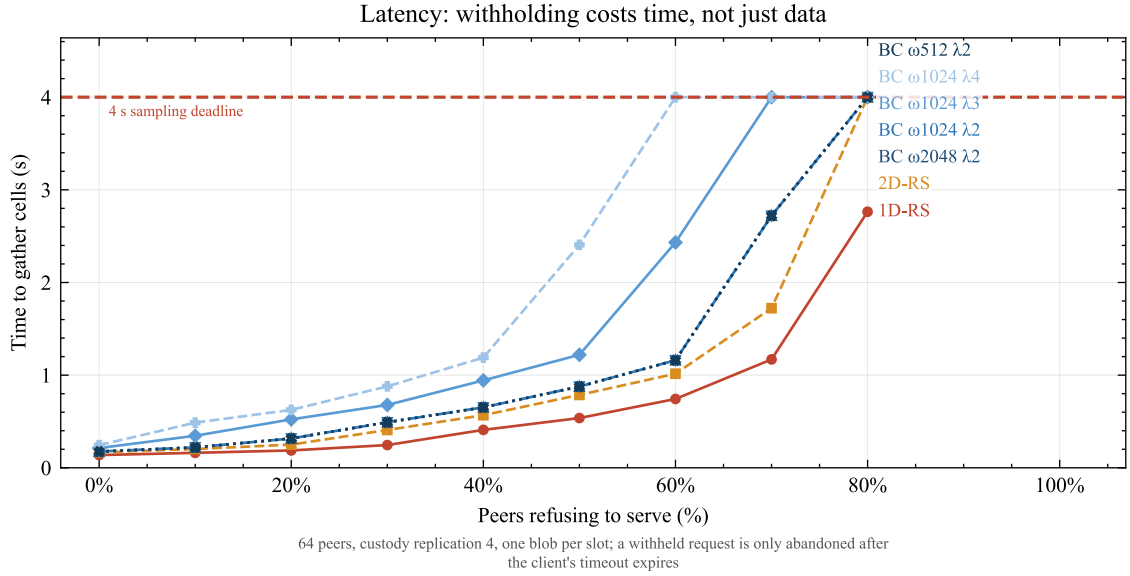
Decode times come from this project's Python/`galois` implementation and are, by a rough estimate rather than a measurement, two to three orders of magnitude slower than production field arithmetic. Absolute figures describe *this code*; comparisons between schemes are the transferable part. `--speedup N` divides decode by N to ask what production arithmetic would look like.



Custody replication, not the code, is what absorbs peer failure. A cell is lost only when every one of its four custodians withholds, so the erasure the code actually sees is roughly the fourth power of the fraction of bad peers. The dashed line is that p⁴ prediction, computed from the model and not fitted; the solid line is what 30 trials per point produced. From 20% withholding up, measurement sits slightly below the line, because the network is finite: exactly round(64p) peers withhold and custodians are drawn without replacement, so the chance that all four are bad is the hypergeometric C(A,4)/C(64,4), a little under p^4 — the peer-count figure below shows that effect directly. At 10% both are effectively zero. The practical reading: 40% of peers withholding costs about 2% of the data.



What each scheme survives when peers withhold. No erasure rate is injected here — losses come from peers refusing to answer, a whole cell at a time, and every trial is decoded for real. BC ω1024 λ4 breaks first: 70% withholding yields 0.233 erasure and it recovers 17 of 30, with ω1024 λ3 close behind at 28 of 30. At 80% withholding (erasure 0.400) the λ2 rows outlast 2D-RS — ω1024 λ2 recovers 18 of 30 and ω512 λ2 14, against 2D-RS's 7 — the reverse of the coding benchmark's order, where 2D-RS tolerates more than any block-circulant row. The difference is what is lost: the benchmark erases scattered symbols, the network whole cells. 1D-RS, being MDS, depends only on how many symbols are missing and recovers every trial. 30 trials per point.



Withholding costs the victim time, not only data. A peer that never answers is not free to route around: the client has to burn its request timeout before it can fall back to another custodian, so gather time climbs with the adversarial fraction even at rates where every cell is eventually retrieved. This is the propagation-latency axis the brief asks for, and it is invisible in any figure measured only at zero adversaries. The schemes separate by how many cells they need: BC ω1024 λ4 tolerates the least erasure, so it must fetch the most, so it is the first to hit the four-second deadline — at 60% withholding — while 1D-RS, needing the fewest cells, never reaches it in this sweep (2.8 s at 80%).

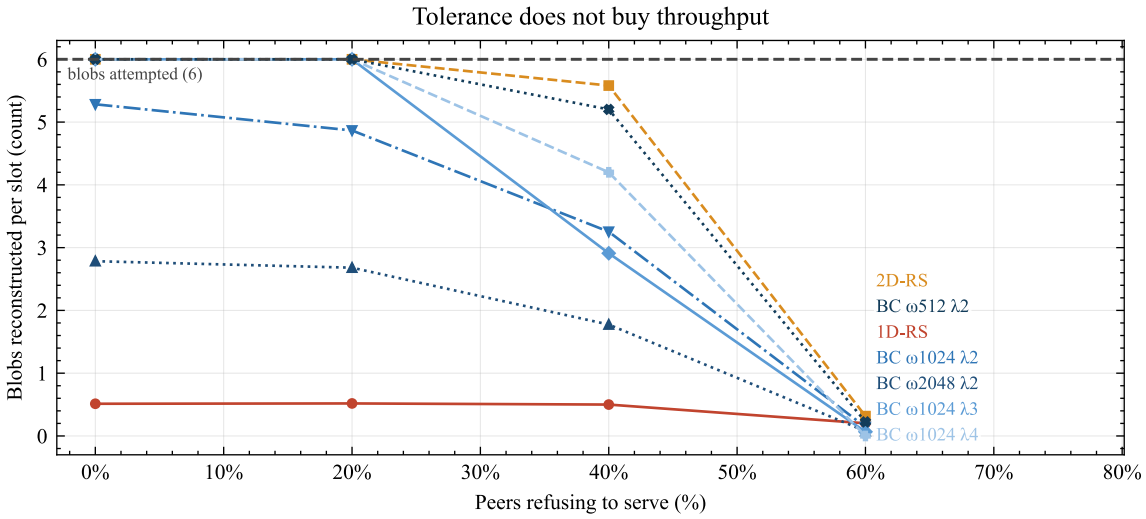
One blob per slot, 50 Mbps (selected rates; full sweep in runA.json)

scheme	adversarial	recovered	erasure	KiB	gather	decode	total	blobs/slot
1D-RS	0%	30/30	0.000	262	0.14s	0.00s	0.14s	1.00
1D-RS	20%	30/30	0.001	262	0.19s	0.19s	0.38s	1.00
1D-RS	40%	30/30	0.021	257	0.41s	2.17s	2.58s	1.00
1D-RS	60%	30/30	0.114	232	0.74s	9.03s	9.77s	1.00
1D-RS	70%	30/30	0.233	201	1.17s	16.81s	17.98s	0.67
1D-RS	80%	30/30	0.400	157	2.76s	26.08s	28.85s	0.42
2D-RS	0%	30/30	0.000	270	0.17s	0.00s	0.17s	1.00
2D-RS	20%	30/30	0.001	270	0.25s	0.12s	0.37s	1.00
2D-RS	40%	30/30	0.021	265	0.57s	0.95s	1.52s	1.00
2D-RS	60%	30/30	0.114	239	1.02s	1.26s	2.27s	1.00
2D-RS	70%	30/30	0.234	207	1.72s	1.44s	3.16s	1.00
2D-RS	80%	7/30	0.400	162	4.00s	1.80s	5.80s	0.23
BC ω2048 λ2	0%	30/30	0.000	262	0.18s	0.03s	0.21s	1.00
BC ω2048 λ2	20%	30/30	0.001	262	0.32s	0.06s	0.38s	1.00
BC ω2048 λ2	40%	30/30	0.021	257	0.65s	0.52s	1.17s	1.00
BC ω2048 λ2	60%	30/30	0.114	232	1.16s	2.88s	4.04s	1.00
BC ω2048 λ2	70%	29/30	0.233	201	2.72s	6.46s	9.14s	0.97
BC ω2048 λ2	80%	4/30	0.400	157	4.00s	10.55s	14.55s	0.11
BC ω1024 λ2	0%	30/30	0.000	262	0.18s	0.01s	0.19s	1.00
BC ω1024 λ2	20%	30/30	0.001	262	0.32s	0.03s	0.35s	1.00
BC ω1024 λ2	40%	30/30	0.021	257	0.65s	0.27s	0.93s	1.00
BC ω1024 λ2	60%	30/30	0.114	232	1.16s	1.50s	2.66s	1.00
BC ω1024 λ2	70%	29/30	0.233	201	2.72s	3.23s	5.91s	0.97
BC ω1024 λ2	80%	18/30	0.400	157	4.00s	5.91s	9.91s	0.60
BC ω512 λ2	0%	30/30	0.000	262	0.18s	0.01s	0.18s	1.00
BC ω512 λ2	20%	30/30	0.001	262	0.32s	0.02s	0.33s	1.00
BC ω512 λ2	40%	30/30	0.021	257	0.65s	0.15s	0.81s	1.00
BC ω512 λ2	60%	30/30	0.114	232	1.16s	0.82s	1.98s	1.00
BC ω512 λ2	70%	30/30	0.233	201	2.72s	1.78s	4.50s	1.00
BC ω512 λ2	80%	14/30	0.400	157	4.00s	3.18s	7.18s	0.47
BC ω1024 λ3	0%	30/30	0.000	262	0.21s	0.02s	0.23s	1.00
BC ω1024 λ3	20%	30/30	0.001	262	0.52s	0.04s	0.56s	1.00
BC ω1024 λ3	40%	30/30	0.021	257	0.94s	0.35s	1.30s	1.00

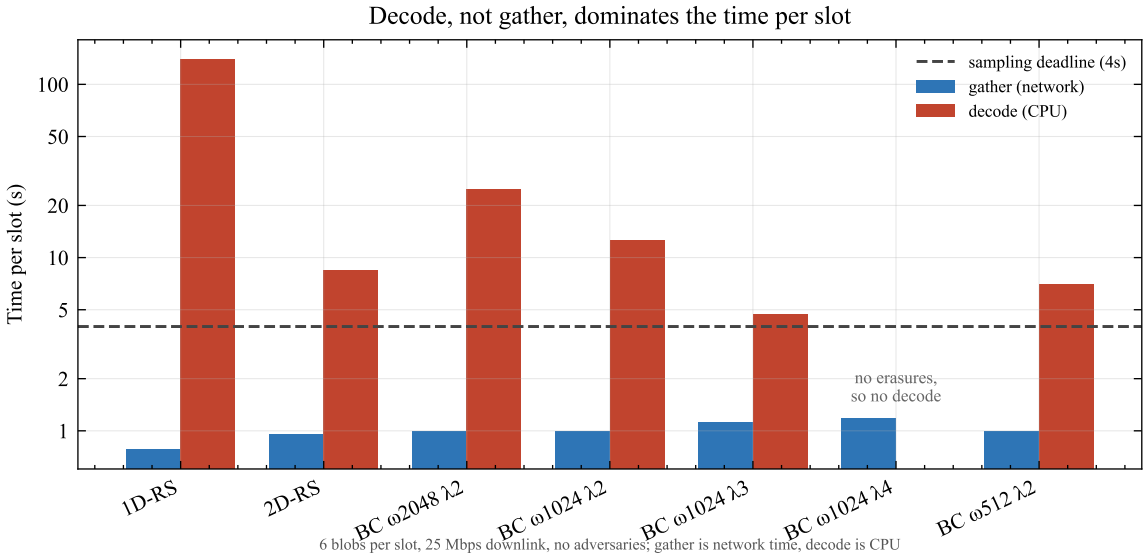
scheme	adversarial	recovered	erasure	KiB	gather	decode	total	blobs/slot
BC ω1024 λ3	60%	30/30	0.114	232	2.43s	1.93s	4.36s	1.00
BC ω1024 λ3	70%	28/30	0.233	201	4.00s	4.09s	8.09s	0.93
BC ω1024 λ3	80%	0/30	0.400	157	4.00s	—	—	0.00
BC ω1024 λ4	0%	30/30	0.000	262	0.24s	0.00s	0.25s	1.00
BC ω1024 λ4	20%	30/30	0.001	262	0.63s	0.01s	0.64s	1.00
BC ω1024 λ4	40%	30/30	0.021	257	1.19s	0.18s	1.37s	1.00
BC ω1024 λ4	60%	30/30	0.114	232	4.00s	1.01s	5.01s	1.00
BC ω1024 λ4	70%	17/30	0.233	201	4.00s	2.00s	6.00s	0.57
BC ω1024 λ4	80%	0/30	0.400	157	4.00s	—	—	0.00

Six blobs per slot, 25 Mbps — where bandwidth starts to bite

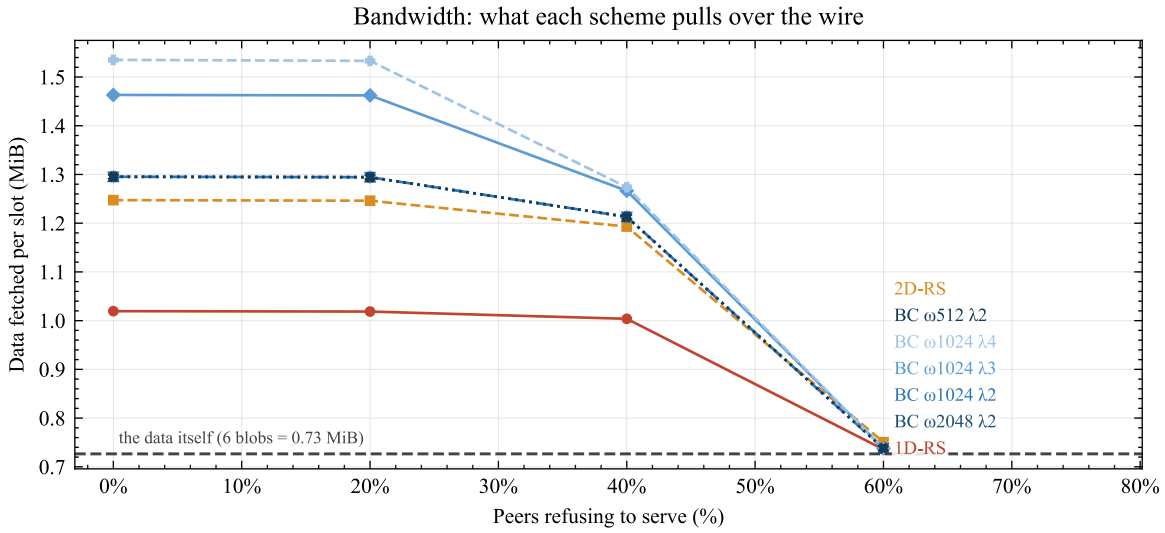
scheme	adversarial	recovered	erasure	KiB	gather	decode	total	blobs/slot
1D-RS	0%	30/30	0.336	1044	0.78s	139.55s	140.33s	0.51
1D-RS	20%	30/30	0.336	1043	1.59s	137.41s	139.00s	0.52
1D-RS	40%	30/30	0.348	1028	3.14s	140.81s	143.95s	0.50
1D-RS	60%	15/30	0.514	753	4.00s	174.35s	178.34s	0.20
2D-RS	0%	30/30	0.212	1277	0.95s	8.40s	9.35s	6.00
2D-RS	20%	30/30	0.213	1276	1.98s	8.41s	10.39s	6.00
2D-RS	40%	29/30	0.248	1222	3.72s	8.76s	12.47s	5.58
2D-RS	60%	2/30	0.517	769	4.00s	11.19s	15.19s	0.32
BC ω2048 λ2	0%	30/30	0.156	1326	0.99s	24.87s	25.86s	2.78
BC ω2048 λ2	20%	30/30	0.157	1325	2.03s	24.83s	26.86s	2.68
BC ω2048 λ2	40%	28/30	0.214	1242	3.79s	34.10s	37.87s	1.77
BC ω2048 λ2	60%	2/30	0.511	756	4.00s	58.71s	62.71s	0.08
BC ω1024 λ2	0%	30/30	0.156	1326	0.99s	12.64s	13.63s	5.28
BC ω1024 λ2	20%	30/30	0.157	1325	2.03s	12.77s	14.79s	4.87
BC ω1024 λ2	40%	30/30	0.214	1242	3.79s	18.34s	22.13s	3.25
BC ω1024 λ2	60%	2/30	0.511	756	4.00s	29.39s	33.39s	0.14
BC ω512 λ2	0%	30/30	0.156	1326	0.99s	7.00s	7.99s	6.00
BC ω512 λ2	20%	30/30	0.157	1325	2.03s	7.02s	9.05s	6.00
BC ω512 λ2	40%	30/30	0.214	1242	3.79s	10.05s	13.84s	5.20
BC ω512 λ2	60%	2/30	0.511	756	4.00s	17.41s	21.41s	0.22
BC ω1024 λ3	0%	30/30	0.047	1498	1.11s	4.67s	5.79s	6.00
BC ω1024 λ3	20%	30/30	0.047	1497	2.26s	4.87s	7.13s	6.00
BC ω1024 λ3	40%	26/30	0.183	1297	3.94s	17.51s	21.44s	2.91
BC ω1024 λ3	60%	1/30	0.511	756	4.00s	32.10s	36.10s	0.07
BC ω1024 λ4	0%	30/30	0.000	1572	1.17s	0.01s	1.18s	6.00
BC ω1024 λ4	20%	30/30	0.001	1570	2.37s	0.08s	2.44s	6.00
BC ω1024 λ4	40%	21/30	0.179	1304	3.97s	7.70s	11.66s	4.20
BC ω1024 λ4	60%	0/30	0.511	756	4.00s	—	—	0.00



Throughput at six blobs per slot and 25 Mbps — tolerance does not buy throughput. 1D-RS tolerates the most erasure of any scheme here and reconstructs the fewest blobs, about 0.5 of the six, because one 8192-length polynomial division costs about 23 s per blob. 2D-RS, BC ω512 λ2, ω1024 λ3 and λ4 keep up with all six until 20% withholding; at zero withholding ω1024 λ2 manages 5.3 and ω2048 λ2 2.8, its large blocks being the slowest block-circulant decode. Note this is throughput *implied by* the measured time — blobs attempted, scaled by the measured recovery rate and by the share of the work that fits in the 12s slot — not an independently measured rate.

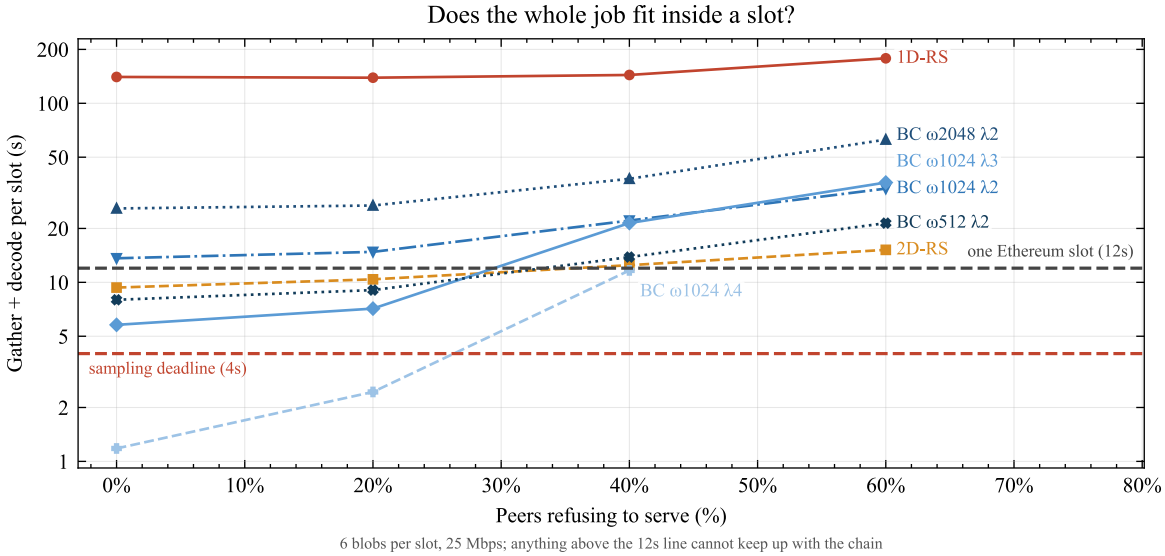


Where the time goes, six blobs, no adversaries, log scale. Gather and decode are drawn as separate bars rather than stacked, because stacking on a log axis would misrepresent the sum. Every scheme gathers in about one second; decode runs from about 5 s to 140 s above that. This is the central result of the network layer: apart from the one scheme that avoids decoding altogether, arithmetic rather than bandwidth or latency decides whether a scheme fits the slot. BC ω1024 λ4 is that exception, and it shows no decode bar because it has the weakest decoder: it needs 81% of the codeword, and with the over-request margin and the requests already in flight it ends up holding all of it, so there is nothing to reconstruct (the decoder returns in a few milliseconds). It buys its speed with bandwidth.



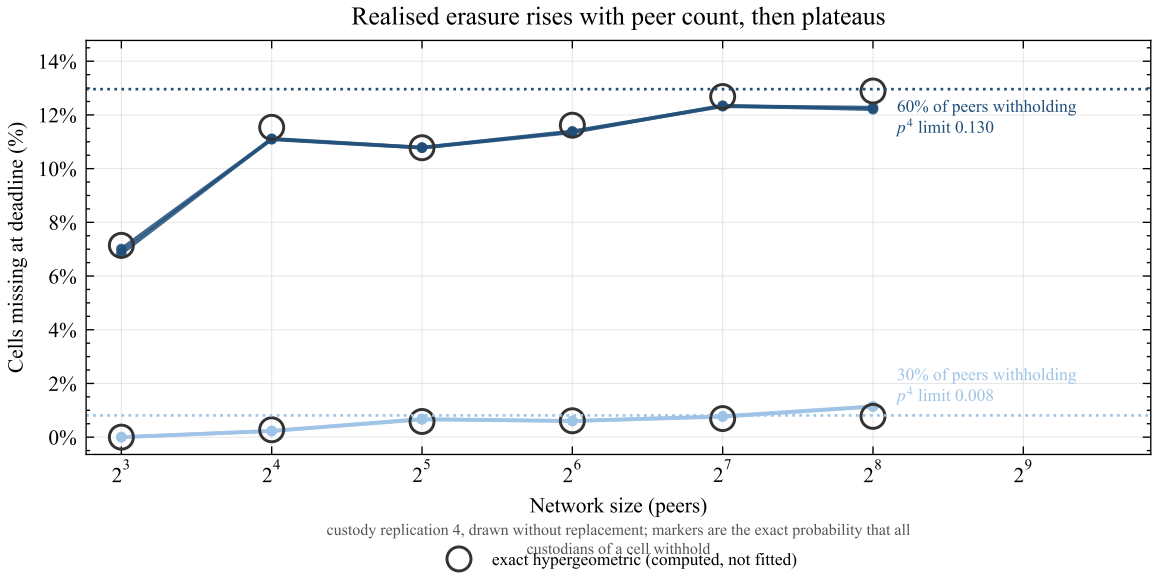
6 blobs per slot, 25 Mbps downlink, 64 peers. the fall at high withholding is cells that never arrived, not a saving

The bandwidth a node actually pulls. The download target is set from each scheme's measured tolerance — the same quantity behind the coding benchmark's surviving-data figure — so the ordering here follows that figure by construction. What the simulation adds is the real cost on top: the over-request margin, requests already in flight when the target is met, and anything re-requested after a timeout. 1D-RS fetches least (1.02 MiB for six blobs, against 0.73 MiB of actual payload) and BC ω1024 λ4 most, at 1.54 MiB, the whole codeword. Read the fall at high withholding carefully — it is cells that never arrived, not a saving.



6 blobs per slot, 25 Mbps; anything above the 12s line cannot keep up with the chain

The question the whole network layer exists to answer. A scheme is only usable if gather plus decode finishes inside an Ethereum slot. At six blobs and zero adversaries 2D-RS (9.3 s), BC ω512 λ2 (8.0 s), ω1024 λ3 (5.8 s) and λ4 (1.2 s) are under twelve seconds, with ω1024 λ2 just over at 13.6 s; by 40% withholding only λ4 still averages under twelve seconds, at 11.7 s, and it recovers in only 21 of 30 trials. 1D-RS is not close at any setting — 140 s at zero adversaries, roughly twelve slots' worth of work for one slot's data. This is the figure that reframes the comparison: erasure tolerance is not the binding constraint, decode throughput is.



Growing the network makes things slightly worse, up to a limit. At 60% withholding and fixed replication 4, custodians are drawn without replacement, so in a small network it is combinatorially hard for all four custodians of a cell to be adversarial — with 8 peers only 5 are bad and there are few ways to choose 4 of them. As the network grows this constraint relaxes and the loss rate climbs toward the independent-sampling limit $p^4 = 0.130$. The open markers are the exact hypergeometric probability, again computed rather than fitted; both schemes sit on it. The lesson for scalability claims is that what protects a cell is the *fraction* of the network custodying it, so peer count cannot be varied without saying what replication does too.

What the results show

Decode dominates; the network barely registers. Gathering takes between 0.14 seconds and the 4-second deadline. Decoding takes up to 174. At every parameter setting tried, except where a scheme avoids decoding altogether, the bottleneck is arithmetic, not the link or the latency. Cutting bandwidth fivefold, from 25 Mbps to 5 Mbps, moved 2D-RS's six-blob total from 9.5s to 10.6s — an 11% change, because the extra second of transfer sits next to 8.5 seconds of decode (a separate 30-trial run at zero withholding, `--schemes 2D-RS --blobs 6 --downlink 5 --real-decode`). At six blobs the spread is starker still: every scheme gathers in about one second, while decode ranges from 0.0s to 139.5s.

That single finding reframes the comparison. The coding benchmark reported decode cost as one column among several; placed in a slot budget it becomes the only column that decides anything.

Losing whole cells is not like losing scattered symbols. The coding benchmark erases symbols one at a time at random; the network loses whole cells of 64 at once. Decoding the network's own erasure patterns shows that this changes who fails. Near the thresholds the block-circulant rows recover more often than their benchmark thresholds predict, and 2D-RS less often: at 80% withholding (erasure 0.400) ω1024 λ2 recovers 18 of 30 and ω512 λ2 14, against 2D-RS's 7 — although in the benchmark 2D-RS tolerates 0.400 and no λ2 row more than 0.349. A likely reason, not tested here, is that whole-cell losses make the load on each block far more uneven, and the block-circulant decoder only needs one block under budget to start. 1D-RS, being MDS, is unaffected: only the number of missing symbols matters to it. Cell-level loss also makes the Reed-Solomon decoders faster. With the same 1,920 symbols missing, 1D-RS decodes in 17.4s when they are whole cells against 23.2s when they are scattered, and 2D-RS in 1.5s against 2.9s.

1D-RS tolerates the most and delivers the least. At 80% of peers withholding it is the only scheme still recovering every trial, 30/30, against 18/30 for ω1024 λ2, 14/30 for ω512 λ2, 7/30 for 2D-RS and 4/30 or fewer for the rest. It needed 28.8 seconds to do it, more than twice the slot. At six blobs it reconstructs

0.51 of them per slot. Best fault tolerance in the study, worst throughput, and the two facts have the same cause: one enormous polynomial division.

Four schemes keep up with six blobs; 1D-RS is nowhere near. 2D-RS, BC ω512 λ2, ω1024 λ3 and ω1024 λ4 reconstruct all six, against ω1024 λ2's 5.28, ω2048 λ2's 2.78 and 1D-RS's 0.51. 2D-RS (8.4s of decode for the six blobs) and ω512 λ2 (7.0s) do it without giving up much tolerance. λ3 and λ4 trade tolerance for speed — λ3 tolerates 0.255 where 1D-RS tolerates 0.499, but decodes roughly 30× faster (4.7s against 139.5s) — and inside a deadline that trade is strongly favourable. ω1024 λ2, decoding 11× faster than 1D-RS, misses the slot by under two seconds.

There is a visible progression inside the three ω1024 rows, and it is a bandwidth-for-compute trade. λ2 fetches 1326 KiB and decodes for 12.6s; λ3 fetches 1498 KiB and decodes for 4.7s; λ4 fetches 1572 KiB and decodes for 0.0s, because its tolerance is so low that it must fetch essentially the whole codeword and then has nothing left to reconstruct. The weakest decoder ends up doing no decoding at all — by paying for it on the wire.

Network-level custody replication absorbs most peer failure before the code is involved. At 70% adversarial peers the realised erasure fraction is only 0.233, because a cell dies only when all four of its custodians do. The erasure code's tolerance is not the binding constraint until the adversarial fraction is very high — which suggests custody replication and code strength are substitutes, and that the cheaper one should be bought first.

More peers is not automatically safer, and can be worse. Sweeping peer count at fixed custody replication produced a result worth stating because it runs against intuition. Custodians are drawn without replacement, so in a small network it is combinatorially hard for all four custodians of a cell to be adversarial: with 8 peers and 60% withholding only 5 peers are bad, and there are just 5 ways to choose 4 of them out of 70. As the network grows that constraint relaxes and the loss rate climbs toward the independent-sampling limit p^4. Measured erasure at 60% withholding runs 0.068 at 8 peers, 0.111 at 16, 0.114 at 64 and 0.123 at 256, against the exact hypergeometric values 0.071, 0.115, 0.116 and 0.129 — agreement within 0.007 at every point. At 30% withholding 2D-RS took 0.33s with 8 peers and 0.83s with 64, entirely because the larger network lost more cells and therefore had more to decode.

The lesson is that what protects a cell is not the peer count but the *fraction of the network custodying it*. Growing the network while holding replication fixed quietly dilutes that fraction. Any scalability claim about node count has to say what happens to replication at the same time, or it is not saying anything.

A weaker decoder costs bandwidth as well as tolerance. BC fetched 1326 KiB where 1D-RS fetched 1044 KiB for the same six blobs, because it needs a larger surviving fraction to decode at all. At equal storage, the scheme with the weaker decoder pays on the wire too — a cost the isolated coding benchmark could not show.

Calibrated mode against real decoding, and two bugs it caught

The simulator's default mode does not run the decoders: it reads decode time and recovery from the coding benchmark's measured curves, so a whole sweep takes seconds rather than an hour. Every run in this report uses `--real-decode` instead. Running the default mode on the same seeds — so the network, and every erasure pattern, is identical and only the decoder differs — shows where that shortcut holds (run A, trials recovered of 30):

scheme	70% withholding: calibrated → real	80%: calibrated → real
1D-RS	30 → 30	30 → 30
2D-RS	30 → 30	18 → 7
BC ω2048 λ2	29 → 29	1 → 4
BC ω1024 λ2	29 → 29	1 → 18
BC ω512 λ2	29 → 30	1 → 14
BC ω1024 λ3	21 → 28	0 → 0
BC ω1024 λ4	5 → 17	0 → 0

In run A every trial up to 60% withholding recovers either way, run C is identical, and run B differs by one or two trials per point. Near the thresholds, though, the calibrated mode is wrong in both directions, because the benchmark's curves were measured on scattered erasures and the network loses whole cells. Its decode times are also too high for the Reed-Solomon schemes, which decode cell-aligned losses faster (above); for the block-circulant rows they agree to within a few percent at realistic erasure. The calibrated mode stays useful for exploring, but the numbers reported here come from real decodes.

Two errors surfaced this way and are fixed:

1. **Zero-erasure rounds invented work.** The cost curve was held flat at its lowest *measured* rate (0.02), so a round with nothing missing was charged several seconds of decoding. `--real-decode` reported 0.00s and exposed it.
2. **Gather time was reporting the deadline back.** The first version ran the environment to the deadline and reported `env.now`, which is the deadline by construction. It now records the moment the node actually held enough cells to start decoding.

Limitations

One topology. The brief asks for "standard network architectures", plural. This models a flat network with uniform random custody. Structured overlays, clustered or geographically correlated latency, and churn are not represented. This is the largest remaining gap.

One-hop fetches only. A node requests cells directly from custodians. There is no gossip propagation, no forwarding, no upload-side cost — so this measures a node's *download* experience, not network-wide load.

Decode cost is Python-speed. Stated above and worth restating: absolute timings are, by rough estimate, two to three orders of magnitude off production, which is what `--speedup` exists for. Relative comparisons hold.

The download target is calibrated. How much a node fetches is set from the coding benchmark's measured tolerance — the highest erasure rate at which every benchmark trial recovered. A node needs some fetch policy, and this is the one place the benchmark's numbers still enter these runs; it is also why the bandwidth ordering follows the benchmark's surviving-data figure.

One blob decoded per slot. In run B the node gathers six blobs, but the decoder runs on the first blob's erasure pattern and its time is counted six times. The other five blobs lose different cells, so their outcomes could differ trial by trial.

Statistical treatment is lighter than the coding benchmark's. 30 trials per point with no confidence intervals, against the benchmark's Wilson intervals and per-scheme trial budgets. Adequate for the size of the effects being reported here — a 12× throughput gap does not need a confidence interval — but it should be tightened before any close call is claimed.

Running it

```
pip install simpy galois numpy matplotlib

python3 network_sim.py                                  # defaults
python3 network_sim.py --blobs 6 --downlink 25          # bandwidth pressure
python3 network_sim.py --peers 8,16,64,256              # sweep network size
python3 network_sim.py --real-decode --trials 3         # ground truth
python3 network_sim.py --speedup 100                    # production arithmetic
python3 network_sim.py --figures figs.md                # SVG, docs-styled
python3 network_sim.py --json results.json              # machine-readable
python3 network_sim.py --no-visuals                     # table only
```

`--peers` and `--adversarial` both accept comma-separated lists and are swept as a cross product with the scheme list, so a single invocation can cover scheme x network size x adversarial pressure.

The three committed runs are exactly these, decoding every trial for real. Runs A and B take about an hour each and run C about twenty minutes (without `--real-decode` each takes seconds, but see the comparison above):

```
S=1D-RS,2D-RS,BC-1024-L2,BC-1024-L3,BC-1024-L4,BC-2048-L2,BC-512-L2
python3 network_sim.py --schemes $S --adversarial 0,0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8 \
    --peers 64 --blobs 1 --downlink 50 --replication 4 --trials 30 --real-decode --json
docs/runA.json --no-visuals
python3 network_sim.py --schemes $S --adversarial 0,0.2,0.4,0.6 \
    --peers 64 --blobs 6 --downlink 25 --replication 4 --trials 30 --real-decode --json
docs/runB.json --no-visuals
python3 network_sim.py --schemes 2D-RS,BC-1024-L2 --adversarial 0.3,0.6 \
    --peers 8,16,32,64,128,256 --blobs 1 --downlink 50 --replication 4 --trials 30 --real-
decode --json docs/runC.json --no-visuals
```

What it draws

Every run prints three terminal views beneath the results table, because the table alone buries the one thing that matters:

- **Throughput** — a bar per configuration, blobs reconstructed per slot.
- **Where the time goes** — gather and decode as one stacked bar per configuration, with a marker showing where the 12-second slot falls. This is the figure that makes the central finding obvious at a glance: the network segment is a sliver and the arithmetic segment is everything. A segment that rounds to zero width is still drawn one cell wide, so "too small to see" stays distinguishable from "absent".
- **Recovery grid** — scheme x peer count against adversarial fraction, marked ● all trials recovered, ◐ some, ○ none.

`--figures out.md` writes the first two as inline SVG in the same style as the rest of `docs/`, so they can be pasted into the write-up directly.

The figures in *this* report are not those, and are not drawn by hand either. They are generated from the committed run data by `make_figures.py`:

```
python3 make_figures.py --only netA netB netC
```

`docs/runA.json` holds the adversarial sweep at one blob, `runB.json` the same sweep at six, and `runC.json` the peer-count sweep. Regenerating from those files rather than redrawing means a figure cannot drift out of step with the numbers quoted beside it, and the hypergeometric and p^4 curves are computed in the plotting code rather than transcribed.

Scheme names match `benchmark_results.json`, so every parameterisation the coding benchmark measured can be driven through the network layer without touching either file.